

ENVIRONMENTAL SOUND ALERT MODULE

25-26J-223

Project Proposal Report (Draft)

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Department of Information Technology

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DECLARATION

I declare that the work presented here is original and that we have not plagiarized in any way, shape, or form any portion of this proposal into another work that has already been submitted for a degree or diploma in any other university or higher educational institution without acknowledgement.

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The supervisor and co-supervisor should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor



Date

29/8/2025

Signature of the co-supervisor



Date

28/08/2025

Abstract

This research aims to develop an innovative solution to improve communication and safety for deaf individuals, particularly in Sri Lanka, by combining multiple assistive technologies. The primary focus is on creating a real-time, context-aware SSL communication system, which integrates hand, face, and body gesture recognition with an emotion-aware voice synthesis system. The system also includes a bidirectional communication module, enabling seamless interaction between deaf and hearing individuals through avatars and speech-to-sign translation. In addition, the research addresses a crucial challenge faced by deaf individuals—the inability to perceive critical environmental sounds, such as alarms, sirens, and announcements. To tackle this, the study proposes the Environmental Sound Alert Module, which utilizes real-time sound classification and edge computing to detect and prioritize urgent sounds. The system ensures low-latency, privacy-respecting processing by performing sound classification directly on the device using CNNs. Furthermore, it provides multimodal notifications (visual, vibration feedback, and on-screen messages) to ensure immediate awareness of critical sounds. This study investigates the feasibility of these integrated components, focusing on real-time processing, urgency-based prioritization, and the user experience. The system will bridge existing gaps in assistive technologies by offering a more efficient, secure, and context-aware solution for deaf users, significantly enhancing their safety and communication in various environments such as homes, public spaces, and transportation hubs. This research will lay the groundwork for improving situational awareness and ensuring inclusive communication for deaf individuals, ultimately offering them greater independence and a safer, more interactive environment.

Keywords: *Environmental Sound Alert Module, real-time sound classification, edge computing, multimodal notifications, urgency-based prioritization.*

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LIST OF ABBREVIATIONS

Sinhala Sign Language	SSL
Convolutional Neural Networks	CNN
Artificial Intelligence	AI
Mel-Frequency Cepstral Coefficients	MFCC
User Interface	UI
Graphics Processing Unit	GPU

1. INTRODUCTION

Deaf individuals face significant challenges in their daily lives due to their inability to perceive environmental sounds, such as alarms, sirens, and other auditory cues that signal potential danger. The importance of assistive technology in addressing these challenges has been well-documented in prior research. Studies have focused on sound recognition systems that provide alerts for deaf individuals in home, healthcare, and public environments. However, these existing solutions have several limitations, such as latency, reliance on cloud-based processing, and lack of real-time urgency-based prioritization. For example, Yağanoğlu and Kose (2018) developed a wearable device to recognize critical sounds like fire alarms and transmit alerts via vibration [1]. Similarly, Tariq et al. (2019) used a speech emotion detection systems to help elderly individuals by recognizing emotional cues and sending real-time alerts [2].

To successfully develop this project, a strong understanding and expertise in several key areas are required, including:

1. **Environmental Sound Recognition:** This involves using machine learning techniques, particularly Convolutional Neural Networks (CNNs), to classify environmental sounds in real-time.
2. **Edge Computing:** Mastery of edge computing techniques is necessary to ensure that sound classification and alert systems are processed on-device, reducing latency and ensuring privacy. Edge AI frameworks like TensorFlow Lite will be used for real-time processing.
3. **Urgency-based Prioritization:** Developing a model to assess the urgency of detected sounds and trigger the appropriate alert mechanism, such as visual or tactile feedback, is essential for the real-time aspect of this system.
4. **Multimodal Notification Systems:** The development of an intuitive interface that integrates visual, vibration, and on-screen feedback for alerts will be critical for user interaction.

Currently, most environmental sound recognition systems for deaf individuals rely on cloud-based solutions to detect and classify sounds. While these systems can offer accurate sound detection, they suffer from high latency due to the need to transmit data to the cloud for processing. This can result in delayed alerts that compromise safety. Edge

computing has emerged as a potential solution to this problem, enabling on-device processing and ensuring faster responses. However, most current systems still focus on basic sound classification without considering the urgency of the sound or providing personalized notifications. Thus, there remains a significant gap in creating privacy-respecting, real-time systems that prioritize sound alerts based on their urgency.

Several researchers have attempted to solve similar problems by developing sound recognition systems tailored for deaf users. For example, Mete Yağanoğlu and Camal Kose (2018) created a wearable device to alert users to critical sounds via vibration [1]. While these solutions address the detection of individual sounds, they often fall short in prioritizing alerts or ensuring real-time, privacy-respecting operation. Moreover, the solutions are typically cloud-dependent, which introduces concerns about latency and privacy.

This project will build upon previous work by introducing real-time, on-device sound classification to eliminate latency issues and ensure privacy. Unlike existing systems, this approach will incorporate an urgency-based prioritization mechanism, which will ensure that the most critical sounds (e.g. alarms, sirens) are detected and trigger immediate alerts. Additionally, the use of edge computing will enable faster processing and eliminate the need for cloud data transmission, addressing the privacy concerns associated with current systems. By integrating these novel features, this system will provide a more reliable and efficient solution for deaf individuals, offering both safety and enhanced situational awareness.

1.1 Background and Literature Survey

Deaf individuals face numerous challenges, particularly when it comes to perceiving environmental sounds that are crucial for their safety and situational awareness. While hearing aids and cochlear implants have helped many individuals with partial hearing loss, complete hearing loss often results in an inability to perceive essential sounds such as alarms, sirens, and announcements. The importance of assistive technologies to address these barriers has been the subject of significant research in recent years.

Technologies designed to aid deaf individuals have primarily focused on visual alerts and vibration feedback. For instance, wearable devices for sound detection and alert the user through vibration [1]. While this system effectively alerts the user to critical sounds, it suffers from the limitation of not prioritizing alerts based on urgency and relying on cloud-based processing, which introduces delays and potential privacy concerns. These limitations have spurred the development of systems with a greater focus on real-time processing and privacy protection.

Chen et al. (2021) demonstrated the potential of 3D audio signal processing in improving sound localization and detection, which could be applied to real-time systems for deaf users [3]. This approach could enhance the detection and localization of environmental sounds, particularly in complex and dynamic environments like public spaces or transportation hubs, where it is crucial to identify the source of critical sounds such as alarms and sirens.

Sound recognition systems use machine learning techniques to detect and classify sounds in the environment. Deep learning models, particularly Convolutional Neural Networks (CNNs), have become increasingly popular in this domain due to their effectiveness in feature extraction and classification tasks [4]. CNNs are highly efficient in processing complex sound patterns, making them ideal for environmental sound detection. Studies have demonstrated the application of CNNs for sound classification, such as speech emotion detection and environmental sound recognition, offering promising results for real-time, automated systems [5].

While several systems exist for recognizing individual sounds, they often fail to address the need for urgent notifications. For example, systems that detect alarms or sirens do so in isolation, without prioritizing which sounds should be more immediately addressed. This leads to a critical gap in providing deaf users with the necessary context for different sounds they might encounter. Tariq et al. (2019) introduced IoT-based systems for

detecting emotional cues in speech, yet, similarly, these systems fail to integrate sound urgency or offer a comprehensive approach to situational awareness for deaf users [2].

One of the key developments in sound recognition has been the use of edge computing for real-time processing [6]. Unlike traditional cloud-based systems, which process data remotely, edge computing allows devices to process data locally, reducing latency and ensuring privacy. Recent advancements in edge AI frameworks, such as TensorFlow Lite and CoreML, have enabled the deployment of machine learning models directly on mobile and embedded devices, which is particularly useful for applications requiring fast, immediate responses.

MobileNet-EAL, a model discussed by Wang et al. (2020), leverages edge computing for efficient sound recognition, particularly useful in devices with limited computational resources [7]. This approach not only enhances the speed and efficiency of sound detection but also preserves user privacy by processing data locally without needing cloud-based servers.

Edge computing is particularly beneficial in applications for deaf users, as it allows for faster sound classification and immediate feedback without the delays associated with cloud processing. For example, a real-time alert system for detecting sirens or alarms can immediately notify users without the delay that cloud-based systems introduce. By processing data locally, edge computing also ensures that sensitive information, such as audio data, is not transmitted to the cloud, thereby protecting the user's privacy.

Most existing sound recognition systems for deaf individuals treat all detected sounds with equal importance, which fails to capture the urgency of critical sounds like alarms or sirens. Urgency-based prioritization of sounds is an important factor that has been largely overlooked in existing technologies [8]. By integrating context-aware sound classification with real-time prioritization mechanisms, it becomes possible to alert deaf individuals more effectively by highlighting the most pressing sounds.

Recent research has addressed this gap in urgency-based sound prioritization. Li et al. (2020) explored the use of a dual attention mechanism in a residual network, which could enhance the performance of sound classification systems, particularly in environments with overlapping sounds [9]. This technique allows the model to focus on the most relevant sounds, improving the accuracy of sound detection in complex environments.

Previous research has shown that it is possible to integrate multiple notification systems for deaf users, including vibration and visual cues. However, few systems have implemented multimodal notifications in response to specific sound types, especially in real-time. Mete Yağanoğlu and Camal Kose (2018) and other similar studies have demonstrated the success of wearable devices that use vibration as an alert mechanism [1], but these systems do not prioritize alerts according to the urgency of the sound, leading to inefficient use of notifications.

To address these issues, recent work has introduced multitask learning and source separation techniques. For instance, Zhang et al. (2021) focused on source separation for sound event detection, enabling systems to differentiate between overlapping sounds like alarms or sirens in dynamic environments [10]. This could significantly improve the reliability and effectiveness of sound event detection, ensuring that critical sounds are prioritized appropriately.

1.2 Research Gap

While significant advancements have been made in sound recognition systems for deaf individuals, several key gaps remain that need to be addressed to create a truly effective solution. A major limitation of many existing systems is their reliance on cloud-based processing, which introduces latency and raises privacy concerns. Cloud-based systems require transmitting sensitive audio data over the internet for processing, which often leads to delays in sound recognition and alert notifications. In critical situations, this delay can compromise the safety of deaf individuals who rely on immediate alerts. Additionally, there are growing concerns about the security of personal data, as cloud-based systems often store audio recordings that may contain sensitive or private information. My research will focus on developing a real-time, on-device sound detection system that ensures fast responses and maintains privacy through local data processing. This shift to edge computing will reduce latency and eliminate the need for cloud data transmission, offering a more efficient and secure solution for deaf users.

Current systems also fail to address the urgency of different sounds, often treating all detected sounds with equal importance. For example, both fire alarms and background noise might be processed similarly, resulting in missed opportunities to alert users to urgent situations. This lack of urgency-based prioritization reduces the effectiveness of notifications, as deaf individuals may not be immediately alerted to life-threatening sounds such as alarms or sirens. In contrast, non-urgent sounds may trigger unnecessary

or distracting alerts. My approach will integrate urgency-based sound classification, ensuring that critical sounds are detected and immediately prioritized for action, providing context-aware feedback tailored to the urgency of each sound. This will make the system more effective by ensuring that the most important alerts are prioritized and communicated clearly to the user.

Another significant challenge with existing solutions is their limited feedback mechanisms. While some systems use visual alerts or vibration feedback to notify users of detected sounds, these systems often fail to combine multiple forms of notifications to enhance the user's ability to perceive and act on the alert. For example, a system that only provides visual feedback might be ineffective in very bright environments, while a system relying solely on vibration may not be noticed in noisy surroundings. My research will implement a multimodal notification system that uses visual cues, vibration feedback, and on-screen alerts to enhance user awareness, depending on the type of sound and its urgency. This multimodal approach will ensure that users are alerted through the most appropriate channel based on the environment and their preferences, making the notifications more intuitive and effective.

Lastly, while edge computing has been introduced in some sound recognition systems, very few fully leverage this technology to ensure real-time processing and privacy. Most existing systems still depend on cloud computing, which not only compromises speed but also introduces concerns regarding the security of user data. The need for immediate feedback in critical situations is especially important for deaf individuals, and relying on cloud processing delays the delivery of alerts. My research aims to leverage edge AI frameworks to process sound data locally on the device, improving both response times and ensuring that user privacy is maintained. By processing sound data on-device, the system will be able to provide real-time responses while safeguarding sensitive data from being transmitted to external servers.

Through these innovations, my research seeks to fill the existing gaps by developing a comprehensive system that enhances situational awareness for deaf users in dynamic environments. By incorporating real-time sound detection with urgency-based prioritization, the system will ensure that deaf individuals are promptly alerted to critical sounds, such as alarms and sirens, while providing intuitive multimodal notifications. The system will prioritize privacy by ensuring all processing occurs on the device, making it a faster, more secure, and more efficient solution compared to current cloud-based alternatives. This approach will not only improve the safety of deaf individuals but also

contribute to the broader goal of creating more inclusive assistive technologies that can adapt to the diverse needs of users.

1.3 Research Problem

Deaf individuals face significant barriers in perceiving critical environmental sounds, such as alarms, sirens, and public announcements, which are crucial for their safety and situational awareness. The inability to hear these sounds can lead to a lack of awareness about emergencies or hazardous situations, putting deaf individuals at risk. Current assistive technologies, while effective in recognizing some sounds, often fall short in providing real-time, context-aware alerts that prioritize urgent signals. These systems may not respond quickly enough, which can prevent timely intervention in emergency situations.

Additionally, many of the existing solutions rely on cloud-based processing, leading to delays in sound detection and alert notifications. This cloud dependency not only introduces latency but also raises concerns about the privacy of sensitive data, as personal sound recordings are transmitted to external servers. Furthermore, most systems do not consider the urgency of different sounds, treating all detected sounds with equal importance, which means critical alerts might be overlooked or delayed.

The primary problem that this research aims to solve is how to develop a real-time, on-device sound recognition system capable of accurately detecting a wide range of environmental sounds and prioritizing them based on urgency. This system will provide deaf individuals with immediate and context-aware notifications through multimodal feedback, such as visual cues, vibration alerts, and on-screen messages. These notifications will be tailored to the urgency of the detected sound, ensuring that critical events, like fire alarms or sirens, trigger the most prominent alerts while less urgent sounds are treated accordingly.

A significant part of the research will focus on how edge computing can be utilized to process sound data locally, eliminating reliance on cloud systems. This approach will reduce latency, enhance the efficiency of real-time sound processing, and ensure user privacy by keeping all sensitive data on the device. The system will incorporate sound urgency-based prioritization, allowing the most important alerts to be addressed first, enhancing the situational awareness and safety of deaf individuals.

Ultimately, the key challenge is to provide a reliable, efficient, and secure solution that improves the safety and quality of life for deaf individuals. By offering real-time, prioritized notifications without cloud-based delays and respecting privacy, this system will bridge the gap in existing assistive technologies and significantly enhance the user experience.

2. OBJECTIVES

2.1 Main Objective

The primary objective of this research is to develop an integrated assistive system that significantly enhances communication and situational awareness for deaf individuals, particularly in Sri Lanka. The system will combine multiple components to provide a comprehensive solution, including Sinhala Sign Language (SSL) recognition using gestures (hand, face, and body), emotion-aware voice synthesis to reflect the signer's emotional state, and an Environmental Sound Alert Module to detect critical environmental sounds. By integrating these components, the system will enable smooth bidirectional communication between deaf and hearing individuals and provide real-time, context-aware notifications about important auditory cues (e.g., alarms, sirens). This research will address the lack of privacy-respecting, real-time sound detection systems and create a more efficient and secure solution for deaf individuals in dynamic environments like homes, public spaces, and transportation hubs.

2.2 Sub Objectives

1. **Environmental Sound Recognition and Classification:** The goal is to develop a real-time, on-device sound classification system that accurately detects and categorizes critical environmental sounds, such as alarms, sirens, and vehicle horns, with high accuracy. By leveraging machine learning models like Convolutional Neural Networks (CNNs), the system will process sound data locally on the device to reduce latency and ensure privacy. This will enable immediate alerts to users when critical sounds are detected, ensuring a fast and efficient response. The system will be optimized to run smoothly on mobile devices while consuming minimal resources.
2. **Urgency-Based Prioritization:** This objective involves creating a prioritization model that determines the urgency of detected sounds and triggers appropriate alerts based on their criticality. High-priority sounds, such as fire alarms or emergency sirens, will be flagged for immediate attention, while non-urgent sounds will be handled with lower priority. The system will deliver these alerts through multimodal notifications like visual cues, vibration, and on-screen messages. This feature will ensure that the most critical alerts capture the user's attention promptly.

3. **Edge Computing for Privacy and Efficiency:** By implementing edge computing, the system will process all sound data locally on the device, eliminating the need for cloud-based processing. This will significantly reduce latency, allowing for immediate responses, and protect the privacy of the user by ensuring no sensitive data is sent externally. The integration of frameworks like TensorFlow Lite will ensure that sound classification and alert prioritization happen efficiently while minimizing energy consumption and optimizing system performance.
4. **Multimodal Feedback Integration:** This objective is to design an intuitive multimodal feedback system that incorporates visual cues, vibration alerts, and on-screen notifications for deaf users. The feedback will be context-aware, with stronger or more frequent notifications for high-urgency sounds, and subtler alerts for less critical sounds. The system will provide a user-friendly interface that allows for customization, so the user can adjust their preferences for different types of alerts depending on their environment or personal needs.
5. **User Interaction and System Performance:** The aim is to develop a user-friendly interface that enables easy interaction with the system. The interface will allow users to customize their alert preferences, ensuring that the notifications are effective and meet individual needs. Additionally, the system will be optimized for efficiency to ensure minimal battery usage while still providing real-time alerts. The overall system performance will be evaluated for responsiveness, and user feedback will be used to improve its functionality and usability.

3. METHODOLOGY

This study focuses on the development of a real-time Environmental Sound Alert Module designed to enhance situational awareness and safety for deaf individuals by detecting and prioritizing critical environmental sounds. The system integrates machine learning techniques, specifically Convolutional Neural Networks (CNNs), for accurate sound classification and utilizes edge computing to process data locally on the device, ensuring low-latency and privacy-respecting alerts.

Once a sound is detected, the system evaluates its urgency, prioritizing critical sounds like alarms or sirens and triggering immediate multimodal notifications through visual cues, vibration, and on-screen alerts. This methodology ensures that deaf users receive fast, relevant notifications while maintaining privacy by processing all data on-device, without relying on cloud-based systems.

3.1 System Architecture and Workflow

3.1.1 System Diagram

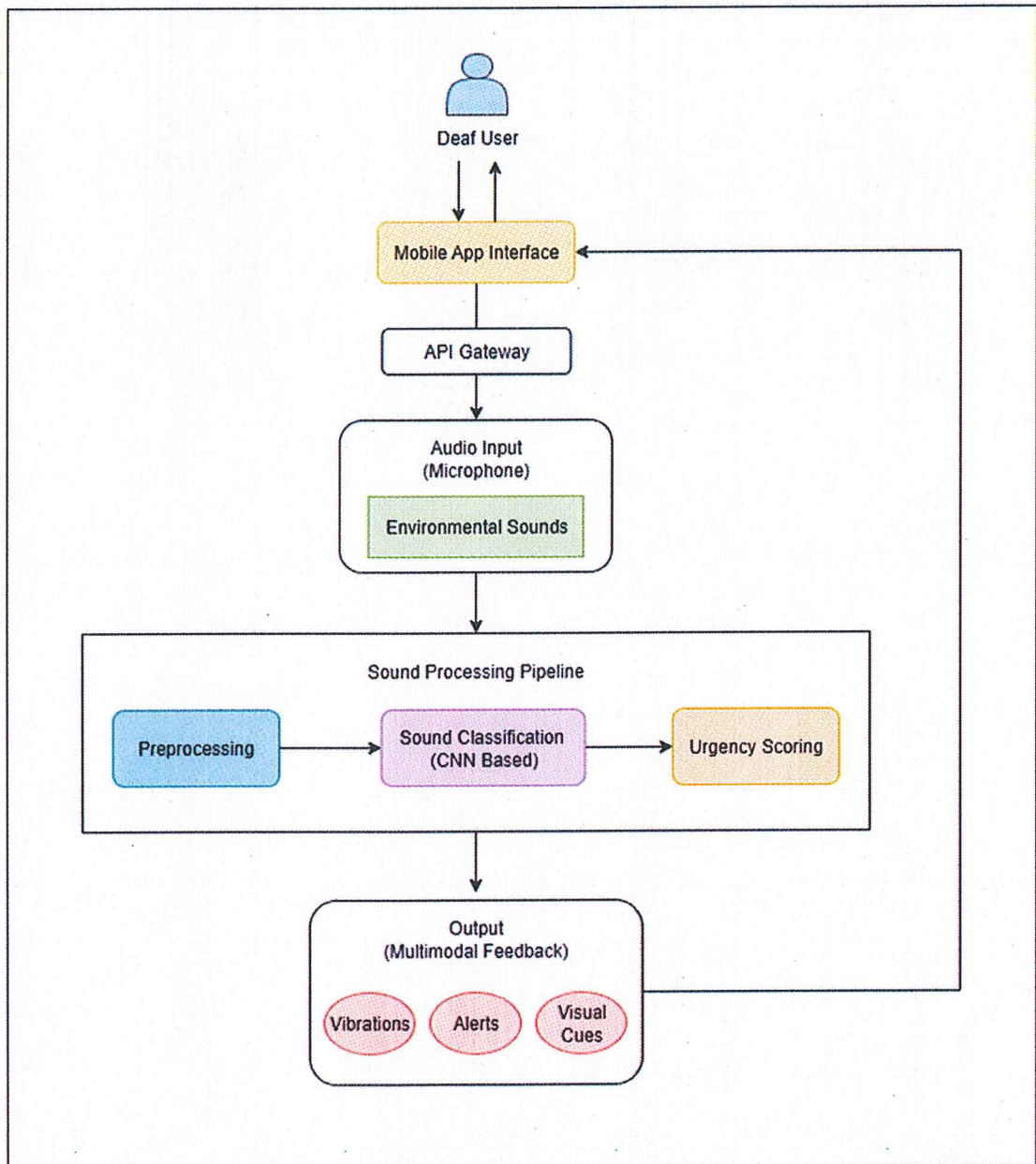


Figure 3-1 System Diagram

3.1.2 System Workflow

1. Audio Capture: Continuously record ambient sounds via the device microphone.
2. Preprocessing: Apply noise reduction, framing, and feature extraction (e.g. MFCC, spectral features).
3. Sound Classification: Feed extracted features into a CNN-based classifier to detect critical alert sounds (e.g. alarms, sirens, horns).
4. Urgency Scoring: Assign urgency levels based on classifier confidence and sound type using a rule-based urgency matrix.
5. Alert Notification: Trigger user alerts - vibration, on-screen banner, flash LED, or emoji animation, according to the urgency score.

3.2 Data Requirements and Collection

- Minimum 10000 labeled audio clips per sound class, recorded across varied noise conditions and device models.
- Continuous field recordings to capture rare events and background noise variability.

The first step in building the sound recognition system involves gathering a diverse set of environmental sound data. This dataset will contain various types of sounds such as alarms, sirens, vehicle horns, glass breaking, and background noise. The data will be collected from a combination of publicly available datasets and real-world recordings to ensure it covers a wide range of environments and scenarios that a deaf user might encounter.

The dataset should include multiple audio clips for each sound category, captured in different environments and at various volumes, so that the model can generalize well to real-world conditions. The data will be annotated with sound labels, and sound events will be tagged with timestamps, specifying the start and end points of each detected sound in the audio files.

3.3 Tasks and Subtasks

1. Data Collection & Labeling

- Gather on-device audio recordings of target sounds (fire alarm, vehicle horn, train announcement, glass breaking, screams) in diverse environments.
- Annotate recordings with sound type labels and occurrence timestamps.

2. Preprocessing Pipeline

- Implement noise reduction (spectral gating) to filter background noise.
- Segment audio into 1-second frames with overlap.
- Extract acoustic features (MFCCs, spectral centroid, zero-crossing rate).

3. Model Development

- Design and train a lightweight CNN classifier on the labeled feature dataset.
- Validate model performance using cross-validation and confusion matrices to ensure $\geq 90\%$ detection accuracy per class.

4. Urgency Scoring Module

- Define urgency levels (High, Medium, Low) for each sound type.
- Implement a rule-based mapping from classifier confidence to urgency level.

5. Mobile Integration & Alert UI

- Optimize and quantize the trained model with TensorFlow Lite for on-device inference.
- Develop UI components for real-time monitoring, including visual alerts (banners, emojis) and haptic feedback (vibration patterns).
- Provide settings to adjust sensitivity, urgency thresholds, and alert modalities.

6. Logging & User Feedback

- Implement local storage (SQLite) for logging detected events and user responses.
- Build a feedback interface to allow users to confirm or dismiss alerts, enabling future model fine-tuning.

7. Testing & Evaluation

- Conduct unit tests on preprocessing, classification, and alert modules.
- Perform end-to-end testing in controlled and real-world environments, measuring detection accuracy, false positive rate ($< 5\%$), and latency (< 200 ms).
- Collect user feedback on alert usability and adjust thresholds accordingly.

3.4 Materials and Resources

- Equipment: Android and iOS smartphones (min. 4 GB RAM, front-facing microphone), laptop with GPU for model training.
- Software: Python, TensorFlow/Keras, TensorFlow Lite, Librosa (audio feature extraction), React Native, SQLite, Git (Version Controlling)
- Audio Data: Custom dataset of environmental sounds recorded in indoor and outdoor Sri Lankan settings; public audio datasets (ESC-50) for augmentation.
- Annotation Tools: Audacity or custom labeling scripts for timestamped annotations.

3.5 Anticipated Outcomes

- A reliable on-device module that detects critical environmental sounds with $\geq 90\%$ accuracy and triggers urgency-based alerts within < 200 ms.
- A user-friendly alert interface combining visual, haptic, and auditory cues to enhance situational awareness for deaf users.
- A logged dataset of real-world sound events to support continuous improvement of the classification model.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- i. **Real-time Sound Detection and Classification:** The system must be able to detect and classify environmental sounds in real-time, such as alarms, sirens, vehicle horns, and other critical sounds, with high accuracy.
- ii. **Urgency-Based Prioritization:** The system should assess the urgency of each detected sound and prioritize alerts based on the criticality of the sound (e.g. fire alarms should trigger immediate alerts, while background noise should be treated as lower priority).
- iii. **Multimodal Notifications:** The system must deliver alerts through multimodal notifications, including visual cues, vibration feedback, and on-screen messages, allowing users to receive context-aware notifications for critical sounds in their environment.
- iv. **Edge Computing for Local Sound Processing:** The system should process all sound data locally on the device using edge computing (e.g. TensorFlow Lite), ensuring real-time processing without reliance on cloud-based services.
- v. **User Customization:** The interface should allow users to customize their preferences, such as selecting the type of notifications (visual, vibration, on-screen), adjusting the intensity of feedback, and defining the urgency levels for different sounds.
- vi. **System Integration:** The system should be designed to integrate seamlessly into the user's daily life, whether at home, in public spaces, or while commuting, with the ability to adapt to various environmental contexts.

4.2 Non-Functional Requirements

- i. **Performance:** The system should perform sound detection and alert prioritization with minimal latency (within milliseconds) to ensure that the user receives immediate alerts for critical sounds.
- ii. **Scalability:** The system should be scalable to handle a wide variety of sound categories (e.g. additional sounds like public announcements or machinery noises) and adapt to different environmental conditions, such as varying noise levels in public spaces or homes.
- iii. **Reliability:** The system must be highly reliable, ensuring that critical sounds are detected consistently, even in noisy or dynamic environments. The notifications must be delivered without failure.

- iv. Privacy: Since the system processes sound data locally on the device, privacy must be always maintained. No sensitive audio data should be transmitted to external servers, ensuring that the user's data remains confidential.
- v. Usability: The system must be user-friendly and intuitive, with a simple interface that allows easy customization of preferences. Deaf individuals with varying levels of technical knowledge should be able to use the system effectively.
- vi. Security: The system should follow best practices for security, ensuring that user preferences and sensitive data are protected from unauthorized access or tampering.
- vii. Compatibility: The system should be compatible with common mobile platforms (e.g. Android and iOS), ensuring accessibility to a wide range of users and devices.
- viii. Maintainability: The system must be easy to maintain, with clear documentation and modular components that can be updated or enhanced as new sound categories or features are introduced.
- ix. Accessibility: The interface and notifications should be accessible to all users, including those with additional disabilities (e.g. visual impairments) by providing options such as contrast adjustment or audio feedback.

5. GANTT CHART

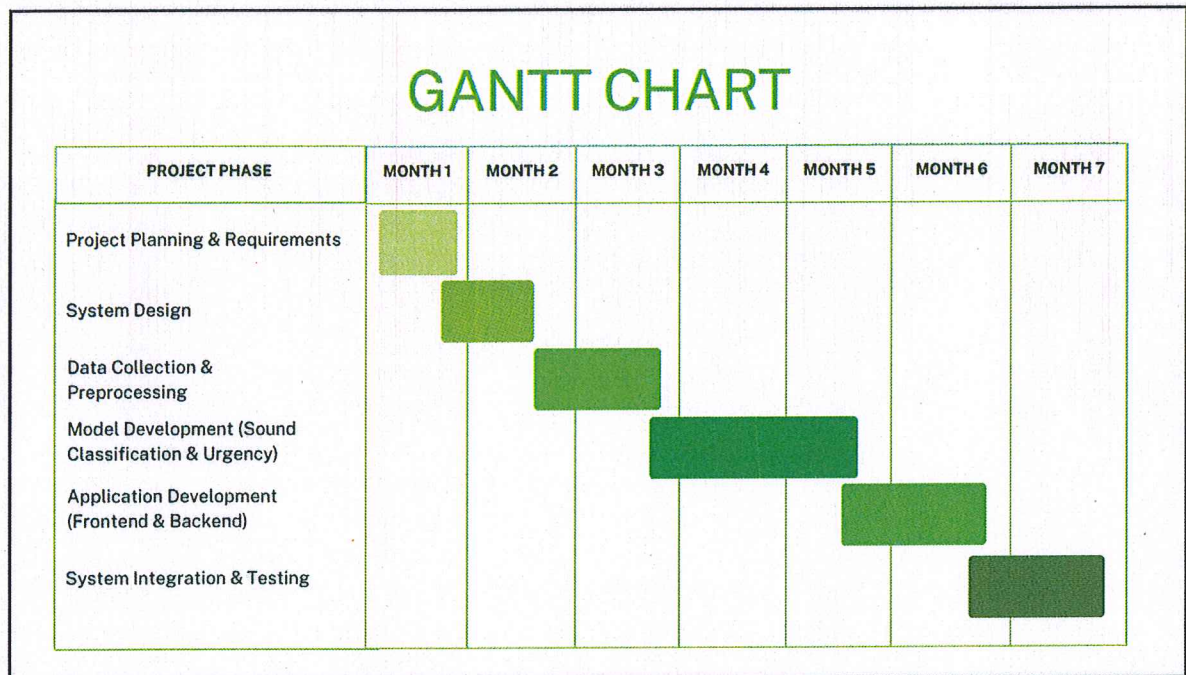


Figure 5-1 Gantt Chart

6. BUDGET AND BUDGET JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Cloud GPU Services (Training)	30,000	For training deep learning models required for sound classification using GPU resources. This will allow the system to process large amounts of audio data quickly and efficiently during the training phase.
Mobile Device for Testing	40,000	To validate the real-time performance of the Environmental Sound Alert Module on mobile devices. This device will be used to test the functionality and efficiency of the system under real-world conditions, ensuring its readiness for deployment.
Miscellaneous	5,000	To cover unexpected expenses, such as additional hardware, software tools, or unforeseen development costs that may arise during the project. This will ensure that the project can continue without delays if unexpected needs occur.

Table 6-1 Budget Estimation Table

Total Estimated Budget: LKR 75, 000

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